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Advanced Software Engineering (ASE)

Design & Construction Principles

Overview, approach, and examination

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1. Introduction

1.1. Overview

In the lecture *Advanced Software Engineering - Design & Construction Principles*, the focus is on current aspects of software engineering and its significance for the Master's program in IT Management.

Advanced Software Engineering - Design & Construction Principles is a module assigned to the technical specialization of the Master's program. The lecture is offered in English, with at least the relevant assignments and presentations to be made or held in English.

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1.2. Learning Outcomes

The module handbook notes the learning outcomes:

*“The students know **advanced topics and interrelationships** in the subject areas of software engineering: Requirements engineering, **specification** as well as **system architecture, development processes** and related aspects of **quality and security**. The students therefore are familiar with **modern principles and paradigms** in the field of software **design, development, deployment and operation**. They can apply this knowledge to practical problems. The analysis, design and development of software systems can be actively accompanied by the students as part of a leading team. They also know how to **coordinate the activities** in the development and deployment chain of **large software systems** and are able to assess technical and economic risks as well as software quality.”*

1.3. Skills

As well as with the skills taught:

“The course covers aspects of the development process from the determination of requirements to quality assurance. Students use common platforms, frameworks and tools to train their ability to plan, monitor and control large complex projects, working on questions in small groups trains in dealing with conflicting goals, promotes discussion, critical faculties and presentation.

The module contributes in particular to the development of leadership competences. The handling of case studies and case studies promotes the necessary decision-making competence. In addition, rhetorical skills and the ability to convince and motivate employees are of great importance.”

1.4. Content

The contents are specified as follows:

- *Requirements engineering*: methods and processes for the definition, documentation and management of functional and non-functional requirements.
- *Software architecture*: design and construction principles, paradigms and structural styles (like microservices), reference architectures, frameworks and libraries.
- *Software development process*: management of complex software development projects, management of software product lines, versioning, prototyping, agile methods.
- *Software deployment, delivery and operating*: Common and crucial aspects of the deployment, delivery and operating chain of software systems as far as these are associated to software engineering: e.g. container, distributed systems, cloud computing, software as a service (SaaS), edge and fog computing

2. Examination

2.1. Expected examinations

The module handbook states regarding the examination performance:
Written examination in the form of a self-directed project including presentation.

The assessment in this module is a written assignment in the form of a self-directed project including a presentation.

2.2. The project: Object and Process

The subject of the project is a **fictional software company**.

For various reasons, the software company needs to restructure its processes and workflows.

- The mit course takes on the task of this restructuring.
- Each student or small groups of up to two students take on the processing of a specified area.

For each topic area, the following elements will be addressed:

a) **Research**

Researching the environment of the specified area.

b) **Decision**

Justified decision for the use of a method, process, tool, or solution approach within the specified area.

c) **Description and Implementation**

Presentation and Implementation (if possible and feasible) of the solution. This can be done exclusively in writing or, for example, with exemplary implementations.

d) **Embedding**

Description of the interaction with the other structural areas by contributing to a solution map.

e) **Status report**

Lecture-accompanying professional presentation of the state of the considerations, progress, and challenges.

f) **Presentation**

Presentation of the results as part of an overarching event.

Students restructure a fictional software company by analyzing and redesigning assigned process areas individually or in pairs. Each project includes research, a justified methodological decision, and the description and (where feasible) implementation of a solution. Results are embedded into an overall solution map and presented through ongoing status reports and a final presentation.

2.3. Artefact 1: Paper (9-12 Pages)

Number of Pages

A term paper is set at 15-20 pages according to the guidelines¹ for academic writing of the HS Mainz. According to the module handbook, the presentation accounts for 40%, while 60% is allocated to the written paper.

This results in a scope of **10 pages** per student for the written paper.

"Pages" refers to content pages, excluding covers, tables of contents, and the like.

The assessment consists of a written paper and a presentation. The written paper accounts for 60% of the grade and the presentation for 40%. Each student is required to submit 10 content pages of written work, excluding cover pages, tables of contents, and similar elements.

Content

The Paper contains a presentation of the results. It includes a description of the developed solution and relevant embeddings:

a) **Research**

From the research area, there may be indications of current practices, the scientific state, sources, or references to conceivable but discarded alternatives.

¹ https://www.hs-mainz.de/fileadmin/Wirtschaft/Services/Pruefungsmanagement/pdf/Downloads/leitfaden/Leitfaden_wissenschaftlicher_Arbeiten.pdf

b) Decision

It presents conceivable options regarding the approach, processes, products, or techniques, as well as a justification for the chosen selection.

c) Description and Implementation

Description of the (proposed) implementation, as well as the concrete integration into the selected company.

d) Embedding

Perhaps a brief mention of the basic embedding – however, this is found on the collaborative map.

The paper presents the developed solution and its integration into the company. It includes research insights, evaluation of alternatives with a justified decision, and a description of the proposed implementation. The overall embedding is outlined briefly, with detailed integration reflected in the collaborative solution map.

2.4. Artefact 2: Presentation (20 Slides, 5 min)

Number of Slides and duration

The presentation is a *Ignite Talk* presentation:

Number of Slides per Person:

20 slides

Time per Person:

5:00 min (corresponding to Ignite Talk with **20 slides * 15 seconds**)

Slide Transition:

Automatic

Special Feature:

In contrast to Ignite Talk, the duration per slide can vary as long as the total remains at 5:00. Extreme example:

First and last slide at 5 seconds, all others around 16 seconds.

Content

The presentation should provide an overview of the work done and elucidate the contents - comparable to the paper. While the paper allows for detailed presentation, the focus of the presentation is to present the results explanatorily, engagingly, and skillfully:

a) Research

The presentation provides insights into the field and the players, the challenges, and typical solutions.

b) Decision

The presentation demonstrates how and why decisions were made.

c) Description and Implementation

The presentation gives an overview of chosen implementations and proposed realizations.

d) Embedding

Perhaps a brief mention of the basic embedding – however, this is found on the collaborative map.

The presentation follows the Ignite Talk format: 20 slides per person, 5 minutes total, with automatic slide transitions; slide timing may vary as long as the total time is maintained.

It provides a concise and engaging overview of the project.

Content includes research insights, decision rationale, proposed implementation, and a brief note on overall embedding (detailed in the collaborative map).

2.5. Artecact 3: Map (1 Page *per course*)

Extent

On one page *per course*, appropriately scaled (A4 with very small font – poster with very large font), the results are presented in a suitable form.

Content

The interaction of the individual solutions becomes clear. Critical transitions may be highlighted – or tools and resources mentioned.

2.6. Accounting for practical implementations

If a student decides to implement a practical application, this may lead to a reduction in the number of pages of the paper.

Simultaneously, this implementation will also be showcased during the presentation: For each page removed from the elaboration, a maximum of 20 seconds will be allocated for presentation.

The removal of pages occurs upon request and in consultation or negotiation with the supervising professor. For this purpose, the specific implementation plan will be outlined.

Example:

A student sets up an environment with corresponding features on Azure DevOps. He receives a credit of 3 pages for this - and additionally, 3 x 15 = 0:45 minutes to showcase the implementation as part of the presentation.

2.7. Submission Deadline

The **submission deadline** was set for <tbid>. This provides a period of <tbid> after our last exchange on <tbid>.

Regarding the technical process, the submission process will be the same as the one you are already familiar with from the RDD. The corresponding link will be provided in a timely manner.